2024-25 年春季学期高等数学 II

思考题 15：矢量微积分

以下题目中的符号可能与书上不同。在就此向我或助教提问之前，请先结合题目说明，尝试猜测一下最合理的理解。

一、以下公式扫描自 J. D. Jackson 的《Classical Electrodynamics》第一页。其中ψ为标量场，$\nabla\psi$表示梯度；$\mathbf{a}, \mathbf{b}$为矢量场，$\nabla\cdot\mathbf{a}$表示散度，$\nabla\times\mathbf{a}$表示旋度。请一一验证这些公式。

$$
\begin{aligned}
\nabla\times\nabla\psi &= 0\\
\nabla\cdot(\nabla\times\mathbf{a}) &= 0\\
\nabla\times(\nabla\times\mathbf{a}) &= \nabla(\nabla\cdot\mathbf{a}) - \nabla^2\mathbf{a}\\
\nabla\cdot(\psi\mathbf{a}) &= \mathbf{a}\cdot\nabla\psi + \psi\nabla\cdot\mathbf{a}\\
\nabla\times(\psi\mathbf{a}) &= \nabla\psi\times\mathbf{a} + \psi\nabla\times\mathbf{a}\\
\nabla(\mathbf{a}\cdot\mathbf{b}) &= (\mathbf{a}\cdot\nabla)\mathbf{b} + (\mathbf{b}\cdot\nabla)\mathbf{a} + \mathbf{a}\times(\nabla\times\mathbf{b}) + \mathbf{b}\times(\nabla\times\mathbf{a})\\
\nabla\cdot(\mathbf{a}\times\mathbf{b}) &= \mathbf{b}\cdot(\nabla\times\mathbf{a}) - \mathbf{a}\cdot(\nabla\times\mathbf{b})\\
\nabla\times(\mathbf{a}\times\mathbf{b}) &= \mathbf{a}(\nabla\cdot\mathbf{b}) - \mathbf{b}(\nabla\cdot\mathbf{a}) + (\mathbf{b}\cdot\nabla)\mathbf{a} - (\mathbf{a}\cdot\nabla)\mathbf{b}
\end{aligned}
$$

二、以下扫描自《Classical Electrodynamics》第二页。其中两个已经在书上有过证明。请一一证明其他公式。提示：（1）可模仿书上对 Gauss 定理和 Stokes 定理的证明（更准确的说，是利用证明中的结论）（2）可利用上一题的结论。

In the following ϕ, ψ, and $\mathbf{A}$ are well-behaved scalar or vector functions, V is a three-dimensional volume with volume element d^3x, S is a closed two-dimensional surface bounding V, with area element da and unit outward normal $\mathbf{n}$ at da.

$$\int_V \nabla\cdot\mathbf{A}\, d^3x = \int_S \mathbf{A}\cdot\mathbf{n}\, da \qquad \text{(Divergence theorem)}$$

$$\int_V \nabla\psi\, d^3x = \int_S \psi\mathbf{n}\, da$$

$$\int_V \nabla\times\mathbf{A}\, d^3x = \int_S \mathbf{n}\times\mathbf{A}\, da$$

$$\int_V (\phi\nabla^2\psi + \nabla\phi\cdot\nabla\psi)\, d^3x = \int_S \phi\mathbf{n}\cdot\nabla\psi\, da \qquad \text{(Green's first identity)}$$

$$\int_V (\phi\nabla^2\psi - \psi\nabla^2\phi)\, d^3x = \int_S (\phi\nabla\psi - \psi\nabla\phi)\cdot\mathbf{n}\, da \qquad \text{(Green's theorem)}$$

In the following S is an open surface and C is the contour bounding it, with line element $d\mathbf{l}$. The normal $\mathbf{n}$ to S is defined by the right-hand-screw rule in relation to the sense of the line integral around C.

$$\int_S (\nabla\times\mathbf{A})\cdot\mathbf{n}\, da = \oint_C \mathbf{A}\cdot d\mathbf{l} \qquad \text{(Stokes's theorem)}$$

$$\int_S \mathbf{n}\times\nabla\psi\, da = \oint_C \psi\, d\mathbf{l}$$